Paper E-01: Universal AFL Synthesis: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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May 2026

Abstract

This synthesis states the common Part IV modelling relation and the release boundary for cross-domain claims. Across Earth systems, engineered systems, socioeconomic systems, domain applications, cosmic and subatomic systems, and abstract or cognitive systems, CDFD/AFL uses the same candidate variables: driving flux Φ, constraint C, responsiveness S, and structural memory M_s. Shared notation does not imply shared material mechanism or empirical validation. The release-local runtime outputs are internal diagnostics and tests that could break the mapping.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, and Λ denotes the Life Number where used. The coefficients α, β, and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution.

1 Common Relation

Part IV uses the operating ratio

$$\Psi_s^* = \left(\frac{\Phi}{C}\right) \cdot S \cdot M_s, \tag{1}$$

with the reduced form Φ/C used only when $S = 1$ and $M_s = 1$. The companion constraint-update form is

$$\frac{\partial C}{\partial t} = \alpha|\Phi| - \beta C + \gamma \nabla^2 C. \tag{2}$$

These equations are a modelling grammar. A domain claim becomes stronger only when each term has a measurable proxy, a time scale, a spatial or network scale, and a failure condition.

2 Cross-Domain Reading

The Part IV sections use the same form for different reasons:

- Part A asks how environmental systems carry heat, water, matter, and biological throughput under constraint.
- Part B asks how engineered systems balance throughput against capacity, state, and responsiveness.
- Part C asks how social and economic systems accumulate memory, friction, and institutional constraint.
- Part D tests additional domain mappings, from immunology and population genetics to information flow.
- Part F applies the notation to cosmic and subatomic examples as hypothesis-level analogies.
- Part G treats language, science, and AI as cognitive or abstract transport systems.

3 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

4 Current Runtime Diagnostic

The diagnostic script `Part_E_Synthesis/supplementary/run_partiv_discovery.py` produces a domain adapter sweep, a universal cascade stress test, two figures, and a generated Markdown summary. The run records finite-value behavior and overload/memory-locking dynamics in the current CDFD Runtime. It is a model record for later measurement, not a claim that the domains have already been validated.

5 Claim Standard

A Part IV claim is stronger when it names the measured Φ, C, S, and M_s proxies; states the expected direction of change; gives the time and network scale; identifies a dataset or experiment; and states what result would falsify the CDFD/AFL interpretation.

6 Boundary

The universal synthesis is not operational advice for infrastructure, finance, medicine, climate intervention, AI safety, or policy. It is a modelling archive with reproducible local diagnostics. Any applied use requires ordinary domain validation and expert review.

7 Conclusion

Part IV presents a single cross-domain modelling language. Its value depends on whether that language produces clearer measurable hypotheses than analogy alone. The release therefore keeps claim status, reproducibility, and falsification standards visible beside the manuscripts and outputs.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs` and `Part_E_Synthesis/figures/`.

8 Reference Layer

References